

Hard-coded pages, SEO, AEO, and GEO

What current Google guidance and 10 academic sources actually support about technology choice, rankings, retrieval, and AI citations.

A hand-coded page is not downranked because it is hand-coded. Search systems evaluate the page they can crawl, render, understand, and trust. The stack matters only when it changes those outcomes.

Bottom line

- Plain HTML, a CMS, server-rendered code, and JavaScript frameworks can all rank.
- Technical SEO is the eligibility layer: access, status codes, rendering, canonicals, links, and metadata.
- Content wins through query and intent alignment, original usefulness, evidence, clarity, and authority.
- Keywords matter as language and topic coverage. The meta keywords tag is ignored, and stuffing is spam.
- AI visibility adds retrieval, reranking, selection, and citation stages. There is no universal GEO hack.

Prepared for Dillon Mohr. This is an evidence synthesis, not a disclosure of any search engine's private ranking system. Google product documentation is used for current implementation rules; academic studies are labeled by venue and scope.

1. The verdict, without the mythology

The claim is directionally correct but needs one important correction. Coding a page by hand does not create a ranking penalty. However, keywords and backend settings are not a substitute for useful content, evidence, reputation, or user value.

The defensible statement: Build with any stack you can operate reliably. Make the result crawlable, indexable, fast, internally connected, semantically clear, and genuinely useful for a defined search intent.

Direct evidence on the code question

- Google says it has no crawling, indexing, or ranking preference among server technologies such as PHP, ASP, CGI, JSP, or CFM, provided the result can be crawled. [G4]
- Google also says code validation does not directly improve ranking, though clean HTML can improve browser compatibility and accessibility. [G4]
- Classic or server-rendered HTML is parsed directly. JavaScript content may require a later rendering step, so server-side rendering or pre-rendering can reduce operational risk. [G3]
- Google's current AI guidance says the same core Search processing supports AI features and that imperfect HTML is understandable. It does not require special AI files or markup. [G1]

What can still go wrong on a hand-coded site

- A 200 page that accidentally carries noindex, a bad canonical, or thin placeholder text.
- Navigation that looks clickable but is not exposed as crawlable HTML links.
- Missing unique titles, descriptions, structured headings, image alternatives, or structured data.
- Orphan pages, duplicate URLs, broken redirects, slow assets, or a sitemap that is never maintained.
- Publishing without a measurement loop in Search Console, analytics, and real conversion data.

2. What actually matters

Search performance is a pipeline. A page must first qualify to be discovered and processed. It then competes on relevance, quality, authority, and usefulness. AI answer systems add more selection stages after retrieval.

Layer	What to optimize	What it does
1. Eligibility	Robots access, public URL, 200 status, indexable text, correct canonical, rendered content	Makes the page available to the system. It does not guarantee ranking. [G2]
2. Meaning	Clear topic, natural query language, descriptive title, headings, entities, internal anchor text	Improves lexical and semantic matching to a user's need. [A1][A5]
3. Usefulness	Original information, complete answers, expert evidence, examples, direct explanations	Supports relevance and quality after eligibility. [G6]
4. Authority	Reputation, independent references, accurate sourcing, meaningful links and mentions	Adds trust and network context. Link structure has been foundational since early web search. [A1]
5. Experience	Fast delivery, mobile usability, accessibility, stable layouts, secure transport	Reduces friction and rendering failure; it cannot rescue irrelevant content. [G1]
6. AI selection	Retrievability, source identity, evidence, concise answer blocks, freshness where relevant	Helps after a source is found, but citation behavior is system-specific and stochastic. [A7-A10]

Keywords: important, but not the old way

- Start with a real question, task, or buying need. Map one primary intent and its necessary subtopics.
- Use the language people use in the title, main heading, opening answer, body, image context, and internal links when natural.
- Cover the topic completely enough to resolve the intent. Modern retrieval includes both lexical and semantic matching.
- Do not use the meta keywords tag as a strategy. Google says it is ignored. [G7]
- Do not repeat terms unnaturally. Google classifies keyword stuffing as spam. [G5]

3. The hard-coded page SEO contract

If these conditions are true, a hand-coded page has no inherent platform disadvantage. Static HTML can even be operationally simpler because the primary content is immediately present. That is a reliability benefit, not a special ranking bonus.

Release gates

Check	Pass condition	Why
Response	Canonical URL returns 200; redirects are intentional; HTTPS works	Basic crawl and index eligibility
Access	Googlebot is not blocked; public resources required to render are available	Crawler can obtain the same meaningful page
Primary content	Useful text is present in initial HTML or reliably server-rendered	Avoids client-only rendering dependencies
Identity	Unique title, H1, description, canonical, author or organization, updated date when relevant	Clarifies page and source meaning
Architecture	Real href links, breadcrumbs where useful, no orphan page, descriptive internal anchors	Supports discovery and topical context
Duplication	One intended canonical; query parameters and alternate routes controlled	Consolidates signals and reporting
Media	Compressed images, dimensions declared, useful alt text, no hidden essential text in images	Performance and accessibility
Structured data	Valid, visible, accurate, and appropriate for the page type	Machine clarity and feature eligibility, not a ranking promise
Measurement	Search Console, analytics, conversions, and a dated query set are configured	Lets evidence replace opinion

Equivalent output means equivalent opportunity. A CMS is not a ranking factor. A framework is not a ranking factor. Their templates, speed, errors, internal linking, duplicate handling, and rendered output can affect the outcome.

4. SEO, AEO, and GEO as one pipeline

AEO means answer engine optimization. GEO means generative engine optimization. The useful distinction is not a new bag of tricks. It is the addition of retrieval, reranking, answer assembly, and citation behavior to the existing search pipeline.

One durable content pattern

- Lead with a direct, accurate answer to a well-defined question.
- Follow with the evidence, method, examples, constraints, and decision detail needed to trust it.
- Use descriptive headings that mirror the page's information hierarchy, not repetitive keyword variants.
- Identify the author or organization, cite original sources, and make dates meaningful and truthful.
- Add structured data only when it matches visible content and the documented page type.
- Connect the page to related authoritative pages through useful internal links.

What the newest academic work changes

- C-SEO Bench found that many conversational-SEO rewrites were ineffective or harmful, while traditional source ranking in the model context remained important. [A7]
- SAGEO Arena evaluated a fuller retrieval-to-generation pipeline and found that body-only optimization can degrade retrieval and reranking; structural information helped. [A9]
- A 252,000-trial SIGIR study found topical relevance and source position were the strongest drivers of first citation in its controlled two-source setting. Formatting-only edits had little consistent effect. [A10]
- The newest critical survey concludes that gains measured after a source is already retrieved do not prove durable organic discovery or traffic gains across platforms. [A8]

What Google currently says

- AI features use the same core Search processing and require normal index eligibility. [G1]
- No special AI text file, machine-readable file, or unique schema is required for inclusion. [G1]
- Google specifically warns against supposed AEO or GEO tricks that do not help users. [G1]

Practical rule: win the earliest weak stage. First make the page discoverable and retrievable. Then improve the answer and evidence. Do not optimize the sentence for citation if the rewrite makes the document less relevant to retrieval.

5. An evidence-led operating plan

The order below keeps engineering, content, and measurement connected.

1. Define demand

- Choose a stable set of real queries grouped by informational, commercial, local, and branded intent.
- Record the audience, expected answer, business outcome, and best page for each query.

2. Establish technical eligibility

- Verify live status, robots directives, rendered content, canonicals, crawlable links, sitemap coverage, and mobile behavior.
- Fix the earliest failing layer before polishing copy.

3. Build the best source page

- Write an original answer, support it with primary evidence, disclose authorship, and include decision-useful detail competitors omit.
- Use the target language naturally in titles, headings, copy, link anchors, and structured fields.

4. Strengthen the graph

- Link related pages intentionally and earn relevant independent references through useful work, not link schemes.

5. Measure by stage

Stage	Primary evidence	Question answered
Crawl	Server logs, URL inspection, robots tests	Can the system fetch it?
Index	Search Console page indexing and canonical data	Was it accepted and consolidated?
Rank	Query, country, device, impressions, average position, landing page	Does it compete for the intended need?
Engage	Qualified sessions, completion, leads, revenue, assisted conversions	Does visibility create value?
AI retrieve	Repeated prompts, variants, engine, date, source list	Is the page selected as context?
AI cite	Citation presence, position, fidelity, volatility across repeated runs	Is it cited accurately and consistently?

Change one meaningful variable at a time. Annotate releases. Compare like-for-like date windows. For AI systems, repeat prompts and paraphrases because one run is not a stable rank. Never treat an isolated citation screenshot as a causal result.

6. Academic sourcebook

Ten academic resources were selected for direct relevance, venue quality, and recency. Nine are peer-reviewed conference publications. The 2026 survey is a preprint and is labeled accordingly.

A1 | Brin, S., and Page, L. (1998)

The Anatomy of a Large-Scale Hypertextual Web Search Engine. Seventh International World Wide Web Conference.

Why it matters: Foundational evidence that web ranking combines text with hyperlink structure and anchor text. It refutes a keyword-only model of search. It is historical context, not a description of today's private algorithm.

Status: Peer-reviewed conference paper | [Open source](#)

A2 | Goren, G., Kurland, O., Tennenholtz, M., and Raiber, F. (2020)

Ranking-Incentivized Quality-Preserving Content Modification. SIGIR 2020. DOI 10.1145/3397271.3401058.

Why it matters: Demonstrates experimentally that content can be modified for rank while explicitly preserving quality and relevance. It supports testing meaningful content changes, not stack-based superstition.

Status: Peer-reviewed conference paper | [Open source](#)

A3 | Mackie, I., Chatterjee, S., and Dalton, J. (2023)

Generative Relevance Feedback with Large Language Models. SIGIR 2023. DOI 10.1145/3539618.3591992.

Why it matters: Shows that generated long-form feedback can improve document retrieval benchmarks. The defensible SEO lesson is semantic and topical coverage, not copying its benchmark percentages into web-ranking promises.

Status: Peer-reviewed conference paper | [Open source](#)

A4 | Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., and Deshpande, A. (2024)

GEO: Generative Engine Optimization. KDD 2024. DOI 10.1145/3637528.3671900.

Why it matters: Formalized GEO and showed content-level visibility gains in a controlled benchmark. The gains are conditional on its experimental setup and do not prove organic retrieval, durable traffic, or cross-engine transfer.

Status: Peer-reviewed conference paper | [Open source](#)

A5 | Ke, Z., Kong, W., Li, C., Zhang, M., Mei, Q., and Bendersky, M. (2024)

Bridging the Preference Gap between Retrievers and LLMs. ACL 2024. DOI 10.18653/v1/2024.acl-long.562.

Why it matters: Establishes that retrievers and generators can prefer different information. It supports treating discovery, retrieval, and answer generation as separate optimization stages.

Status: Peer-reviewed conference paper | [Open source](#)

Academic sourcebook, continued

A6 | Nestaas, F., Debenedetti, E., and Tramer, F. (2025)

Adversarial Search Engine Optimization for Large Language Models. ICLR 2025.

Why it matters: Shows that LLM selection can be manipulated by crafted third-party content. It is a warning against adversarial GEO tactics and evidence that production systems will evolve defenses.

Status: Peer-reviewed conference paper | [Open source](#)

A7 | Puerto, H., Gubri, M., Green, T., Oh, S. J., and Yun, S. (2025)

C-SEO Bench: Does Conversational SEO Work? NeurIPS 2025 Datasets and Benchmarks Track. DOI 10.52202/085713-0923.

Why it matters: Across tasks, domains, and adoption rates, many C-SEO methods were ineffective or harmful. Improving the source's traditional ranking in model context was more effective in the benchmark.

Status: Peer-reviewed conference paper | [Open source](#)

A8 | Martinez, O. (2026)

Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization (2023-2026). arXiv:2607.14035.

Why it matters: Reviews 45 studies and separates retrieval, citation, prominence, and downstream behavior. It finds topical relevance among the more reproducible levers and warns that generic tactics transfer poorly.

Status: Preprint, not peer-reviewed as of this brief | [Open source](#)

A9 | Kim, S., Jeong, W., Kim, S., Lee, S., and Lee, D. (2026)

SAGEO Arena: A Realistic Environment for Evaluating Search-Augmented Generative Engine Optimization. KDD 2026. DOI 10.1145/3770855.3818146.

Why it matters: Evaluates full retrieval, reranking, and generation stages. Existing body-text tactics often degraded earlier stages, while structural information helped. Optimization must be stage-specific.

Status: Peer-reviewed conference paper | [Open source](#)

A10 | Vishwakarma, R., Kumar, S., and Jamidar, R. (2026)

What Gets Cited: Competitive GEO in AI Answer Engines. SIGIR 2026. DOI 10.1145/3805712.3808445.

Why it matters: In 252,000 controlled two-source trials across six LLMs, topical relevance and source position were the largest first-citation drivers. Formatting-only changes had little consistent effect.

Status: Peer-reviewed conference paper | [Open source](#)

7. Current implementation authorities

Academic research explains information retrieval and tests limited systems. Google Search Central is the stronger source for present-tense implementation claims about Google Search.

G1 | Google: AI features and your website

Core Search processing, normal eligibility, no special AI file or schema, no GEO tricks. [Open source](#)

G2 | Google Search technical requirements

Googlebot access, successful HTTP status, and indexable content are minimum requirements, not guarantees. [Open source](#)

G3 | JavaScript SEO basics

How Google crawls, renders, and indexes JavaScript; why server-side rendering or pre-rendering can improve reliability. [Open source](#)

G4 | Crawling and indexing FAQ

No technology preference when output is crawlable; W3C validation is not a direct ranking factor. [Open source](#)

G5 | Spam policies for Google web search

Keyword stuffing, link spam, and other manipulative practices. [Open source](#)

G6 | How Google Search works

Crawling, indexing, and serving results based on many relevance and quality factors. [Open source](#)

G7 | Meta tags and attributes Google supports

Google ignores the meta keywords tag for indexing and ranking. [Open source](#)

Final answer: use hard-coded HTML if it is the clearest, fastest, most reliable way to ship. Do not expect the code choice itself to earn or lose rank. Win through eligibility, intent alignment, original value, evidence, authority, internal architecture, and disciplined measurement.

Evidence limits

Search engines do not disclose complete production ranking functions. Academic papers use bounded datasets, models, prompts, and metrics. Results can establish mechanisms and conditional effects, but they do not justify guaranteed rankings or citations. AI engines are especially volatile; repeat measurements across engines, prompts, dates, and query paraphrases.